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CHAPTER 3 CANONICAL FORMS

Seek a "simple" standard form for matrices, which is invariant under similarity. It will generalize diagonalizability.

§ 3.1 The Jordan Canonical Form (JCF)

Will take time
- set up terminology / logistics
- examine properties & applications
- practice finding JCF & using its features
- Proof: sketch the main parts of the proof.

Important: JCF is a good tool for matrix analysis in a theoretical way. Not amenable to computing.

The main ingredient of JCF is

<u>Definitions</u> A <u>Jordan block</u>, $J_k(\lambda)$ is $k \times k$ upper triangular matrix of the form

$$J_k(\lambda) = \begin{bmatrix} \lambda & 1 & & 0 \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{bmatrix} \in M_k$$

<u>Notice</u>: $\lambda \in \sigma(J_k(\lambda))$, alg. multiplicity of λ is k.

e.g. $J_1(\lambda) = [\lambda]$ (by convention)

$$J_2(\lambda) = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}, \quad J_3(\lambda) = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$$

A <u>Jordan matrix</u> $J \in M_n$ is a direct sum of Jordan blocks:

$$J = J_{n_1}(\lambda_1) \oplus J_{n_2}(\lambda_2) \oplus \dots \oplus J_{n_k}(\lambda_k)$$

$(n_1 + n_2 + \dots + n_k = n)$

$$= \begin{bmatrix} \boxed{J_{n_1}(\lambda_1)} & & & 0 \\ & \boxed{J_{n_2}(\lambda_2)} & & \\ & & \ddots & \\ 0 & & & \boxed{J_{n_k}(\lambda_k)} \end{bmatrix}$$

If $A \in M_n$ is a similar to a Jordan matrix $J \in M_n$, then we call J the <u>Jordan Canonical Form</u> (JCF) of A ($JCF(A)$).

We will see that every $A \in M_n$ has a JCF (that is unique up to the order of $J_{n_1}(\lambda_1), \ldots, J_{n_k}(\lambda_k)$) and in fact every matrix similar to A admits the same JCF.

<u>Initial Observations</u>

(i) $\sigma(J_k(\lambda)) = \{\underbrace{\lambda, \lambda, \ldots, \lambda}_{k\text{-copies}}\}$

(ii) In the definition of a Jordan matrix the scalars $\lambda_1, \lambda_2, \ldots, \lambda_k$ are not necessarily distinct. Nevertheless

$$\sigma(J) = \{\underbrace{\lambda_1, \ldots, \lambda_1}_{n_1 \text{ copies}}, \underbrace{\lambda_2, \ldots, \lambda_2}_{n_2 \text{ copies}}, \ldots, \underbrace{\lambda_k, \ldots, \lambda_k}_{n_k \text{ copies}}\}$$

$= 0$

i.e. $J_m(\lambda) = \lambda I_m$, a contradiction. ∎

(iv) As a consequence of (iii), if any Jordan block in a Jordan matrix J has size ≥ 2, then J is <u>not</u> diagonalizable.

(because direct sum is diag/ble iff every summand is diag/ble).

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Comment on 2.4 #4:

- Main thrust (edition 1):

  $A, B \in M_n$, $AB = BA$, A^{-1} exists.

  Show $A^{-1}B = BA^{-1}$.

or

recall A^{-1} is a polynomial in powers of A. Use that.

$$BA = AB$$
$$\Downarrow$$
$$B = ABA^{-1}$$
$$\Downarrow$$
$$A^{-1}B = BA^{-1}$$

- In 2nd edition:

  "Recall" classical adjoint of $A = [a_{ij}] \in M_n$

$$\operatorname{adj} A = \underbrace{\left[(-1)^{i+j} \det \underbrace{A[\{j\}^c | \{i\}^c]}_{\text{submatrix of } A \text{ when row } j \text{ \& column } i \text{ have been deleted}}\right]}_{\text{cofactors of } A}$$

Example

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$C_{ij} = (-1)^{i+j} \det\left[A[\{j\}^c \mid \{i\}^c] \right]$$

$$C_{11} = (-1)^{1+1} \det \begin{bmatrix} 5 & 6 \\ 8 & 9 \end{bmatrix}$$

$$C_{12} = (-1)^{1+2} \det \begin{bmatrix} 2 & 3 \\ 8 & 9 \end{bmatrix}$$

$\vdots$

Fact: $\boxed{\operatorname{adj} A \cdot A = A \cdot \operatorname{adj} A = \det A \cdot I}$

(p. 22

A^{-1} exists $\Rightarrow \det A \neq 0 \Rightarrow \boxed{A^{-1} = \frac{1}{\det A} \operatorname{adj} A}$

A non-invertible $\Rightarrow \det A = 0 \Rightarrow \operatorname{adj} A \cdot A = 0$

Extra question Show that

$$\boxed{AB = BA \Rightarrow B \operatorname{adj} A = \operatorname{adj} A \cdot B}$$

To show that:

If A is invertible ✓

Otherwise, consider $A + tI$ such that $t \in \mathbb{R}$, $A + tI$ is invertible.

In fact,

$$\det(A + tI) \neq 0$$

for all $t \in (0, \alpha)$ for some $\alpha > 0$.

But then

$$AB = BA$$

$$\Rightarrow (A+tI)B = B(A+tI)$$

so $B(A+tI)^{-1} = (A+tI)^{-1}B$ (by first part)

so $B \operatorname{adj}(A+tI) = \operatorname{adj}(A+tI) \cdot B$

$\forall t \in (0, \alpha)$

Now just let $t \to 0$:

$$B \operatorname{adj}(A) = \operatorname{adj}(A) B$$

Back to JCF:

(v) Look at the eigenspace of a Jordan block

$$J_k(\lambda) = \begin{bmatrix} \lambda & 1 & & 0 \\ & \lambda & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{bmatrix} \in M_k$$

$\sigma(A) = \{\lambda, \lambda, \dots, \lambda\}$

$am(\lambda) = k$

$gm(\lambda) = ?$ $\qquad \dim(N(J_k(\lambda) - \lambda I)) = ?$

<u>Solve</u> $\quad (J_k(\lambda) - \lambda I)x = 0$

$gm(\lambda)$ = dim of the solution set of this homogeneous system.

$$J_k(\lambda) - \lambda I = \begin{bmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{bmatrix}$$

coefficient matrix. $\downarrow$ in echelon form

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{bmatrix}$$

Column 1 has <u>no</u> pivot

All other columns have pivots.

Solving: $x_2 = x_3 = \dots = x_k = 0$

x_1 is free.

i.e.

$$N(J_k(\lambda) - \lambda I) = \text{span}\left\{ \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \right\}$$

So $\text{gm}(\lambda) = 1$.

(vi) Closer look at $J_k(0)$:

(e.g. $J_k(\lambda) - \lambda I = J_k(0)$)

$$J_1(0) = [0]$$

$$J_2(0) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$J_3(0) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Generally (prove inductively), powering up $J_k(0)$, every power "pushes" the band of 1's above the diagonal toward the NE corner of the matrix, until it gets pushed out completely.

Certainly,

$$\left(J_k(0)\right)^k = 0$$

So $J_k(0)$ is a nilpotent matrix.

($J_k(0)$ is <u>not</u> diagonalizable if $k>1$)

Moreover, $\left(J_k(0)\right)^m = 0$, $\forall m \geq k$.

(<u>vii</u>) How about powers of $J_k(\lambda)$, $\lambda \neq 0$.

$$J_1(\lambda) = [\lambda], \quad (J_1(\lambda))^m = [\lambda^m]$$

$$J_2(\lambda) = \begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$$

$$\begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}^2 = \begin{bmatrix} \lambda^2 & 2\lambda \\ 0 & \lambda^2 \end{bmatrix}$$

$$\begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}^3 = \begin{bmatrix} \lambda^3 & 3\lambda^2 \\ 0 & \lambda^3 \end{bmatrix}$$

Guess

$$\begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}^m = \begin{bmatrix} \lambda^m & (\lambda^m)' \\ 0 & \lambda^m \end{bmatrix}$$

(derivative $\frac{d}{dm}$)

$$J_3(\lambda) = \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$\begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix} \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} \lambda^2 & 2\lambda & 1 \\ 0 & \lambda^2 & 2\lambda \\ 0 & 0 & \lambda^2 \end{bmatrix}$$

$$\begin{bmatrix} \quad \end{bmatrix}^3 = \begin{bmatrix} \lambda^2 & 2\lambda & 1 \\ 0 & \lambda^2 & 2\lambda \\ 0 & 0 & \lambda^2 \end{bmatrix} \begin{bmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} \lambda^3 & 3\lambda^2 & 3\lambda \\ 0 & \lambda^3 & 3\lambda^2 \\ 0 & 0 & \lambda^3 \end{bmatrix}$$

$$\begin{bmatrix} J_3(\lambda) \end{bmatrix}^m = \begin{bmatrix} \lambda^m & (\lambda^m)' & \frac{(\lambda^m)''}{2!} \\ & \ddots & (\lambda^m)' \\ 0 & & \lambda^m \end{bmatrix}$$

All entries are derivatives!

Guess general formula.

Main Theorem

Let $A \in M_n$. There exist $S \in M_n$ that is invertible such that

$$A = S \underbrace{\begin{bmatrix} J_{n_1}(\lambda_1) & & 0 \\ & \ddots & \\ 0 & & J_{n_k}(\lambda_k) \end{bmatrix}}_{J} S^{-1},$$

where $n_1 + n_2 + \cdots + n_k = n$. Matrix J is unique up to the order of the Jordan blocks.

Terminology We call J the Jordan Canonical Form of A, $JCF(A)$

Proof Sketch a constructive proof next time.